

Using this course without the internet

Google Colab needs a connection. **Everything else in this course does not.** All the measurements ship inside this repository, and every notebook reads them from the `data/` folder when it is there — nothing is downloaded while you work. The one exception is notebook 6, which calls a live air-quality API.

So the plan is: install Python and seven packages **once, while you have a connection**, then work through the course with the Wi-Fi off.

You will use **JupyterLab**, which shows notebooks in your browser much as Colab does — same cells, same **Shift+Enter**, same charts.

What you need to download

Do this part while you are connected. About **150 MB** in total, once.

What	Size	Where it comes from
The Python installer	~30 MB	https://www.python.org/downloads/
This course (<i>Code</i> ▶ <i>Download ZIP</i>)	~5 MB	the repository's green Code button
Seven Python packages	~110–145 MB	downloaded for you in Step 3

Once installed, the course takes roughly **600 MB** of disk space.

Step 1 — Install Python

Windows

1. Go to <https://www.python.org/downloads/> and download the installer for Windows.
2. Run it, and on the very first screen **tick “Add python.exe to PATH”**. This is the one step people miss, and skipping it causes most of the problems in the troubleshooting list below.
3. Click **Install Now**.
4. Check it worked: press **Start**, type `cmd`, press Enter, and in the black window type `python --version`. You should see a version number.

macOS

1. Go to <https://www.python.org/downloads/> and download the macOS installer.
2. Run it and accept the defaults.

3. Check it worked: open **Terminal** (press ⌘+Space, type `Terminal`) and type `python3 --version`.

Use this Python rather than the one Apple includes with macOS.

Linux

```
sudo apt install python3 python3-venv python3-pip      # Debian / Ubuntu
python3 --version
```

Which version? Anything from **Python 3.11** onwards. This course was last tested on **3.14.3**. If you are preparing several machines for a group, make sure everyone installs the *same* version — see the instructor section at the end.

Step 2 — Get the course files

On the repository page, click the green **Code** button, then **Download ZIP**. Unzip it somewhere you will find again — your Documents folder is fine.

You should end up with a folder containing `notebooks/`, `data/`, `README.md` and `requirements.txt`.

Keep the notebooks inside the `notebooks/` folder. They find the measurements by looking one folder up, so a notebook moved elsewhere will try to download the data instead — and fail when you are offline.

Step 3 — Install the seven packages

Open a terminal in the course folder:

- **Windows** — open the folder in File Explorer, click in the address bar at the top, type `cmd`, and press Enter.
- **macOS** — open Terminal, type `cd` (with a space), then drag the course folder onto the Terminal window and press Enter.

Then create a private workspace for the course and install into it.

Windows:

```
python -m venv .venv
.venv\Scripts\activate
python -m pip install pandas matplotlib plotly requests jupyterlab scipy PyWavelets
```

macOS / Linux:

```
python3 -m venv .venv
source .venv/bin/activate
python3 -m pip install pandas matplotlib plotly requests jupyterlab scipy PyWavelets
```

The download takes a few minutes. When it finishes, your prompt starts with `(.venv)` — that is how you know the workspace is switched on.

What is a “virtual environment”? The `.venv` folder is a private box holding this course's packages, so they cannot disturb anything else on your computer. If your setup ever gets tangled, delete the `.venv` folder and run these three lines again.

There is also a `requirements.txt` in the folder. It installs the same seven packages plus two more (`boto3`, `contextily`) that are only needed for regenerating the course data from scratch — a much bigger download that students do not need.

No internet at all on this machine? See **Preparing machines with no connection** at the end.

Step 4 — Start JupyterLab

In the same terminal, with `(.venv)` showing:

```
jupyter lab
```

Your browser opens with a file browser on the left. Open the `notebooks` folder and double-click `01_welcome_python_basics.ipynb`. Run each cell with **Shift+Enter**, exactly as the notebooks describe.

To stop: close the browser tab, then press **Ctrl+C** twice in the terminal.

Every day after that, three lines from the course folder:

```
cd path/to/tutorial-air-quality-data-analysis
.venv\Scripts\activate           # Windows
source .venv/bin/activate        # macOS / Linux
jupyter lab
```

What is different from Colab

The notebooks were written for Colab, so a few menu names differ. Everything else — cells, Shift+Enter, the  /  self-checks — behaves the same.

The notebook says	In JupyterLab
<i>Runtime ▶ Restart session</i>	Kernel ▶ Restart Kernel
<i>Runtime ▶ Run all</i>	Run ▶ Run All Cells
<i>File ▶ Save a copy in Drive</i>	Ctrl+S / %S — saves straight into the notebook file
the  Files panel	the file browser on the left
“Run anyway” warning	does not appear

Your work is saved in the notebook file on your own computer. There is no Drive copy to make and nothing to lose when you close the tab. If you later want a fresh, unfilled copy of a notebook, download the ZIP again.

The one notebook that needs the internet

Notebook 6 (Bonus: the OpenAQ API) fetches live measurements from the internet and needs a free [OpenAQ API key](#). Do that one while you are connected, or skip it — nothing later depends on it.

Notebooks **1–5, 7 and 8 run completely offline**, including all the maps and charts. The Accra map in notebook 7 is a picture that ships with the course, so no map service is contacted.

If something goes wrong

'python' is not recognized... (Windows) — Python was installed without “Add python.exe to PATH”. Either re-run the installer and tick that box, or type `py` instead of `python`.

No module named pandas — the workspace is not switched on. Your prompt should start with `(.venv)`; if it does not, run the `activate` line again from the course folder.

The browser did not open — look in the terminal for a line starting with `http://localhost:8888/lab?token=...` and paste that whole line into your browser.

Port 8888 is already in use — JupyterLab is already running in another window. Close it, or let JupyterLab pick the next port when it offers.

A notebook tries to download data, or says `FileNotFoundError: ../data/...` — the notebook has been moved out of the `notebooks/` folder. Put it back, or keep the `data/` folder one level above it.

Charts do not appear, or a cell complains about a name it does not know — run the cells in order from the top of the notebook. Each one builds on the last. **Kernel ▶ Restart Kernel and Run All Cells** fixes almost everything.

The interactive chart in notebook 5 is blank — restart the kernel and run the notebook again from the top. The other charts in that notebook are ordinary images and are unaffected.

pip fails behind a company firewall or antivirus — use the offline route below, prepared on a machine that is not restricted.

For instructors: preparing machines with no connection

If each machine can get online once

Walk the group through Steps 1–4 above. Budget about 15 minutes per machine, most of it waiting for the download, and have the course ZIP on a USB stick so 30 people are not pulling it over the same weak connection.

If the machines never get online

Build a USB stick containing the Python installer, the course ZIP, and a folder of pre-downloaded packages (a “wheelhouse”).

On a connected machine **of the same operating system and the same Python version** as the target machines:

```
pip download pandas matplotlib plotly requests jupyterlab scipy PyWavelets -d wheels
```

Then on each offline machine — install Python, unzip the course, and run:

```
python -m venv .venv
.venv\Scripts\activate
python -m pip install --no-index --find-links D:\wheels ^
    pandas matplotlib plotly requests jupyterlab scipy PyWavelets
```

(`--no-index` tells pip not to look for the internet at all. On macOS or Linux use `python3`, `source .venv/bin/activate`, and the path to the folder on the stick.)

Measured on macOS: **109 MB** of packages, giving a 573 MB installed environment.

Building the Windows stick from a Mac

This works, with one trap worth knowing. `pip download` will fetch the Windows versions of the packages, but it still decides *which* packages are needed based on the machine it is running on — so it silently leaves out three packages that only Windows needs, and JupyterLab will not start without them. Ask for them explicitly:

```
pip download --only-binary=:all: --platform win_amd64 --python-version 3.13 \  
pandas matplotlib plotly requests jupyterlab scipy PyWavelets -d wheels-win  
  
pip download --only-binary=:all: --platform win_amd64 --python-version 3.13 \  
pywin32 pywinpty colorama -d wheels-win
```

That produces **143 MB** across 107 files. The `--python-version` must match the Python your students install (3.13 in the example), because these files are tied to a specific version — so tell the room one version and stick to it.

Caveat: that bundle was assembled and checked on a Mac, not installed on Windows. Do one trial install on any Windows laptop before the workshop.

What was tested

On 2026-09-22, on macOS, in a newly created environment: Python 3.14.3 with pandas 3.0.6, matplotlib 3.11.2, plotly 7.1.0, scipy 1.18.1, PyWavelets 1.8.0 and JupyterLab 4.6.4. Notebooks 1–5, 7 and 8 each ran top to bottom reading the local `data/` folder, and JupyterLab served the course folder without incident.